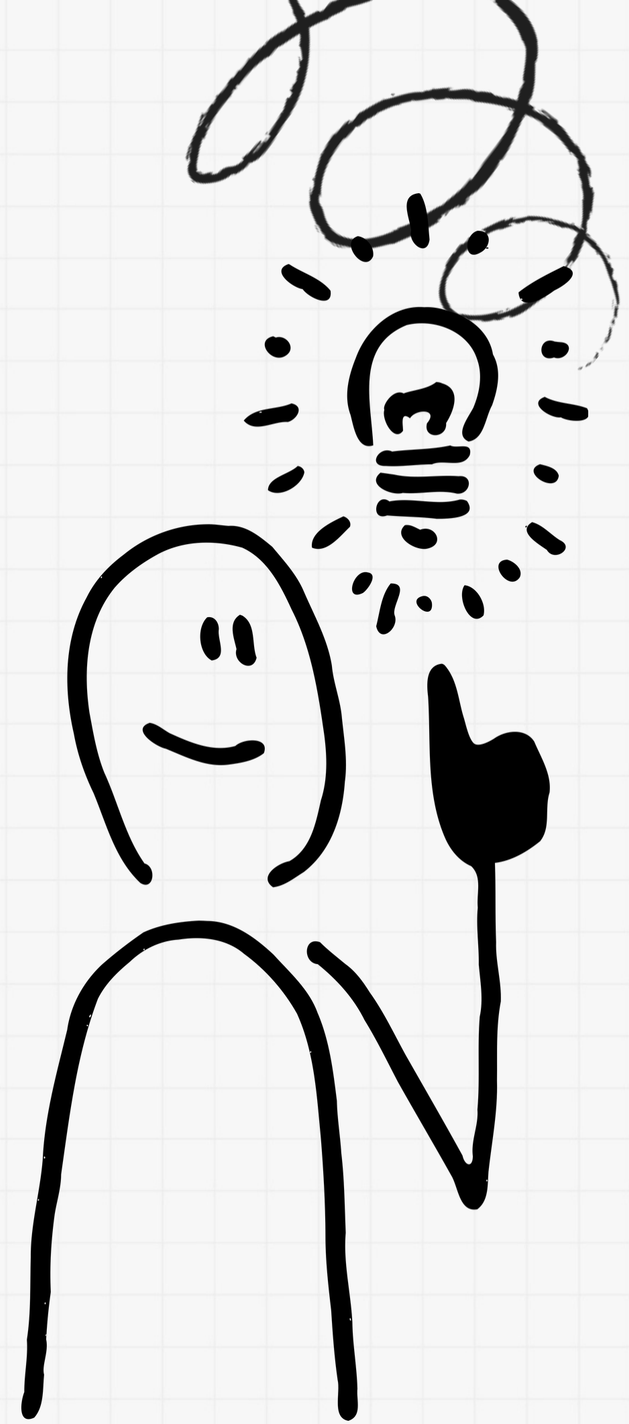
# Self-RAG

## Learning to Retrieve, Generate, and Critique through Self-Reflection

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CS613 · Paper Presentation

Presented by: [Team member names]

# LLMs make things up

- **Knowledge lives in the weights.** It is fixed after training, has no source and is hard to update.
- **Rare facts are the worst.** Few training examples, so probability spreads over plausible but wrong answers.
- **Users cannot verify** an answer that comes with no evidence.
- **The standard fix is Retrieval-Augmented Generation (RAG):** look up passages, then answer. But RAG brings problems of its own.



Why not just trust the model?

no retrieval

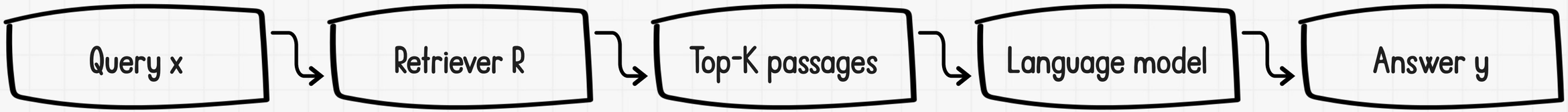
with retrieval

**14.7%** ⇨ **45.7%**

Llama2-13B accuracy on rare-entity questions (PopQA long-tail: entities with fewer than 100 Wikipedia views a month)



# RAG in one slide




dense search over
~21M Wikipedia passages

d1 ... dK
(K = 5 here)

## Dense retrieval

$$\mathrm{sim}(q, d) = \langle E(q),\ E(d) \rangle$$

Embed query and passage separately, rank by dot product. Contriever is trained with a contrastive loss; Self-RAG keeps it frozen.

## Classic RAG (Lewis 2020)

$$p(y|x) \approx \sum_{z \in \mathrm{top}K} p_\eta(z|x)\, p_\theta(y|x, z)$$

The passage is a latent variable: average the answer over passages, weighted by the retriever. Always retrieves.

## In-context RAG (common)

$$y \sim p_\theta(\ \cdot\ |\ [d_1; \ldots; d_K;\ x])$$

Paste all K passages in front of the prompt. No training needed. This is the 'Ret-ChatGPT' baseline.



# Why plain RAG falls short



## Always retrieves

Top-K passages for every prompt, even 'write an essay about your summer'. Wastes compute and derails creative or reasoning tasks.

Ret-ChatGPT on PubHealth:

70.1 → 54.7



## Irrelevant passages

The retriever's top hit can be off-topic, and LMs are easily distracted by irrelevant context (Shi et al. 2023).

Alpaca-13B on PubHealth:

55.5 → 51.1 with retrieval



## No grounding check

Nothing makes the output follow the passages, and nothing checks it afterwards, so citations are unreliable.

Open RAG models on ASQA:

2–8% citation precision

Earlier fixes each cover one part: FLARE retrieves when confidence is low, SAIL always prepends passages, Toolformer learns API calls, RLHF needs expensive PPO, and self-refine needs many prompt calls.

# Core idea: reflection tokens

Add special tokens to the vocabulary and train ONE model, with ordinary next-token prediction, to write them between its own sentences. **Critique becomes next-token prediction.**

### Retrieve

Should I fetch evidence for the next sentence?

**Yes · No · Continue**

sees: x and y<t
before each sentence

### IsRel

Is passage d relevant to the question x?

**Relevant (best) · Irrelevant**

sees: x, d
after the passage

### IsSup

Is every claim in my sentence backed by d?

**Fully (best) · Partially · No support**

sees: x, d, yt
after the sentence

### IsUse

Is the whole answer useful? (not about truth)

**5 (best) · 4 · 3 · 2 · 1**

sees: x, y
once, at the end

# What the model actually writes

**Q: when did Walking Dead season 7 come out?**

[Retrieve=Yes]  ...season 7 premiered on October 23, 2016... 
[IsRel=Relevant] October 23, 2016 [IsSup=Fully] [IsUse=5]

**Q: write a poem about the sunset**

[Retrieve=No] The stars begin to twinkle bright... [Retrieve=No] They flicker like a silent dance... [IsUse=5]

## Why this order?

- **Retrieve first:** decide before writing
- **IsRel before the sentence:** the judgment shapes the answer
- **IsSup after:** check the finished sentence
- **IsUse last:** a property of the whole answer
- ... passages come from the retriever, not the model

## One token stream = chain rule

$$p_{\mathcal{M}}(\tilde{y}|x) = \prod_{t=1}^{T} p(r_t^{ret}) \left[ p(r_t^{rel})\, p(y_t)\, p(r_t^{sup}) \right]^{\mathbf{1}[ret_t]} p(y_t)^{1-\mathbf{1}[ret_t]} \cdot p(r^{use})$$

# Training: distil a critic, then teach the generator

**GPT-4 labels** → **Critic C** → **Annotate 150k** → **Generator M**

Few-shot prompts per token type: 12.6k Retrieve, 19.3k IsRel, 11.2k IsSup, 3.8k IsUse

Llama2-7B fine-tuned on those labels. Agrees with GPT-4: 94% Retrieve, 94% IsSup, 80% IsRel, 74% IsUse

For each sentence: C decides retrieval, R fetches, C labels IsRel / IsSup, keeps one passage. Text unchanged

Llama2 7B / 13B + new tokens, plain next-token training (3 epochs, 4×A100)

**At inference: only M + the retriever**

C and GPT-4 are only used to build the training data. M has learned to be its own critic.



# The two training objectives

## Critic · Eq. 1

$$\max_{C} \; \mathbb{E}_{((x,y),r)\sim\mathcal{D}_{critic}} \log p_C(r \mid x, y)$$

A classifier written as language modelling: predict GPT-4's label r as text.

## Generator · Eq. 2

$$\max_{\mathcal{M}} \; \mathbb{E}_{(x,y,r)\sim\mathcal{D}_{gen}} \log p_{\mathcal{M}}(y, r \mid x)$$

$$\mathcal{L} = -\sum_i m_i \log p_{\mathcal{M}}(\tilde{w}_i \mid x, \tilde{w}_{<i}), \quad m_i = 0 \text{ inside } \langle p\rangle \ldots \langle /p\rangle$$

Writes the answer AND its critique, interleaved.

### Why mask the passages?

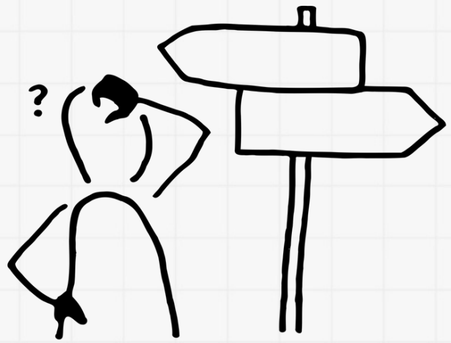
- They are inputs from the retriever, not model output
- Predicting them would teach memorising Wikipedia, or making up passages
- The model still attends to them, so it learns to use them

### Why not RLHF / PPO?

- PPO: about 4 models in memory and on-policy sampling
- Self-RAG: 1 model, static corpus, cross-entropy loss
- Behaviour is changed later at inference, without retraining

# Inference, one sentence at a time

$$\text{retrieve} \Leftrightarrow \frac{p(\text{Yes})}{p(\text{Yes}) + p(\text{No})} > \delta$$




Probability of 'Yes' within the Yes/No group. Default δ = O.2: retrieve even when 'No' is more likely. δ = O for ASQA, where citations are required.

STAGE 01 → STAGE 02 → STAGE 03 → STAGE 04 → STAGE 05

### Retrieve?

Check the threshold above. If no: write the sentence like a normal LM

### Fetch K

R returns the top 5 passages for (x, previous sentence)

### K drafts

One candidate sentence per passage, in parallel, each after its IsRel token

### Critique

IsSup for each (d, sentence) pair; IsUse for usefulness

### Keep best B

Segment-level beam search, B = 2, ranked by the score f

# Scoring a candidate sentence

$$f(y_t, d) = p(y_t \mid x, d, y_{<t}) + w^{Rel} s^{Rel} + w^{Sup} s^{Sup} + w^{Use} s^{Use}$$

$$s^{Rel} = \frac{p(\text{Relevant})}{p(\text{Rel}) + p(\text{Irrel})}$$

$$s^{Sup} = \frac{p(\text{Fully}) + 0.5\, p(\text{Partially})}{p(\text{Fully}) + p(\text{Partially}) + p(\text{No})}$$

$$s^{Use} = \sum_{i=1}^{5} u_i \frac{p(i)}{\sum_j p(j)}, \quad u = (-1, -\tfrac{1}{2}, 0, \tfrac{1}{2}, 1)$$

- **Defaults:** w = (1, 1, 0.5)
- **Expected value** of each critique, in a fixed range
- **Hard mode:** drop any candidate with IsSup = No support

## Worked example

(illustrative numbers)

| | d1: states named after people | d2: baby names in Texas |
|---|---|---|
| Fluency p(yt) | 0.60 | 0.70 |
| s Rel | 0.90 | 0.20 |
| s Sup | 0.80 | 0.25 |
| s Use | 0.57 | 0.33 |
| Score f | 2.59 (chosen) | 1.31 |

Likelihood alone would pick d2, a fluent sentence built on an off-topic passage. The critique terms pick the grounded d1.

# Why it works

## Decomposition

4 small, separately supervised decisions instead of one hard one

## Retrieve if useful

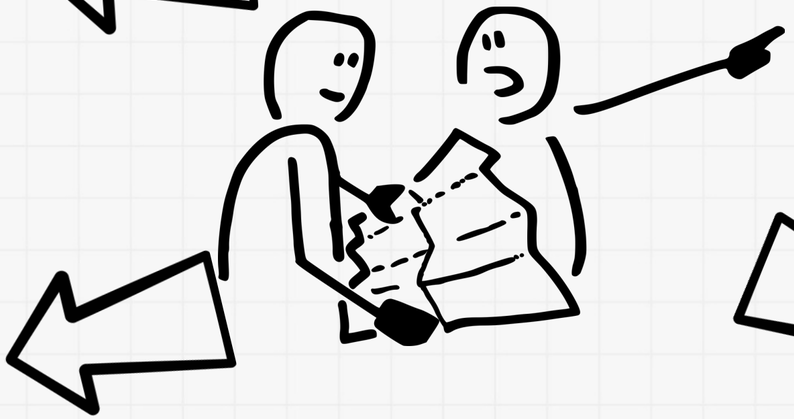
Essays and maths stay a normal LM, so versatility is kept

## Generate + verify

Checking a sentence against one passage is easier than writing a correct one

## Cheap critique

Learned offline with SFT: no RL, no reward model at test time



## One passage each

No mixing of noisy passages: bad ones are scored down

## RLHF in disguise

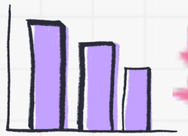
Like $\pi^* = \pi_{ref} \cdot \exp(\text{reward} / \beta) / Z$: changing w changes the reward, no retraining

# Experimental setup



## Closed-set

- PubHealth: health-claim fact check
- ARC-Challenge: science multiple choice
- Metric: accuracy



## Short-form QA

- PopQA long-tail: 1,399 rare-entity questions
- TriviaQA: 11,313 questions
- Metric: gold answer appears in the output



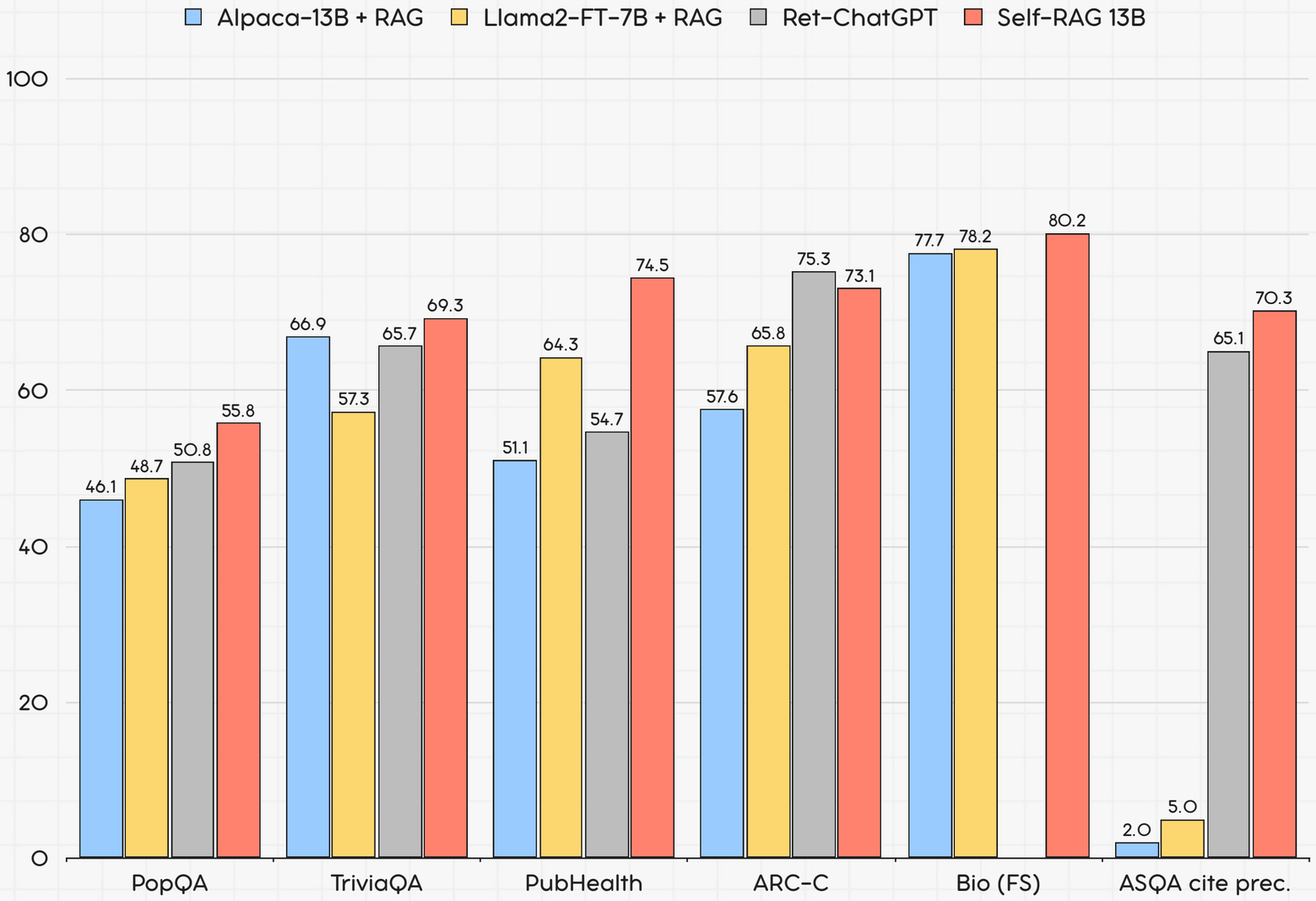
## Long-form

- Biographies: FactScore (% of atomic facts supported)
- ASQA: correctness, fluency (MAUVE), citation precision and recall

**Baselines** No retrieval: Llama2, Alpaca, Llama2-chat, ChatGPT, CoVE-65B · Standard RAG: the same models + the same passages · Same data, no reflection: Llama2-FT · Retrieval-trained: SAIL, Toolformer

**Settings** Llama2 7B / 13B · 150k training pairs · Contriever-MS MARCO top-5 (+5 web results for Bio and QA) · zero-shot · δ = 0.2, B = 2, w = (1, 1, 0.5)

# Main results




- **Best open model** on all six tasks
- **Beats ChatGPT** on PopQA, PubHealth, Bio and citation precision
- **Not everywhere:** ChatGPT without retrieval still wins TriviaQA (74.3) and ARC (75.3)
- **Same data is not enough:** Llama2-FT trails, so the gain comes from the framework

Source: Table 2. Ret-ChatGPT has no Bio score.

# Ablations: every piece matters

| 7B, 50k training examples | PopQA | PubHealth | ASQA em |
|---|---|---|---|
| Self-RAG (full) | 45.5 | 73.5 | 32.1 |
| Train: no retriever | 43.6 | 67.8 | 31.0 |
| Train: no critic | 42.6 | 72.0 | 18.1 |
| Test: never retrieve | 24.7 | 73.0 | - |
| Test: hard threshold | 28.3 | 72.6 | - |
| Test: always top-1 passage | 41.8 | 73.1 | 28.6 |
| Test: drop IsSup score | 44.1 | 73.2 | 30.6 |

- **No critic → ASQA collapses:** reflection tokens are the core
- **No retrieval hurts PopQA, not PubHealth:** needs differ by task
- **Hard threshold retrieves too rarely:** δ is a crucial knob

## Test-time knobs (no retraining)

### w Sup 0 → 2

citation precision 69.7 → 70.5, but MAUVE fluency 96 → 86

### δ up

fewer retrievals; costs PopQA much more than PubHealth

### 5k → 150k data

PopQA about 34 → 55 and still rising

### Human check

IsRel judged correct 90–95%, IsSup 85–90%

# Critical analysis

## Strengths

- No RL: SFT + reranking
- Tunable at test time (δ, w)
- Per-sentence citations with self-labels
- Strong controls: Llama2-FT, ablations

## Limitations

- Every label traces back to GPT-4
- Critic errors become training targets
- Frozen retriever: detects bad passages, cannot fix them
- Up to $B \cdot K = 10$ drafts per step; latency not reported

## Assumptions

- One sentence ← one passage
- GPT-4 judgments match human ones
- The corpus contains the facts
- Token probabilities can simply be added

## Improvements

- Re-query when every IsRel is low
- Learn or calibrate w on a dev set
- Let a sentence use several passages
- Preference training (DPO) on critique pairs

# Impact

**2020**

## RAG

Lewis et al.: retrieval as a latent variable, always on

**2023**

## Self-RAG

Retrieval on demand + self-critique tokens (ICLR 2024)

**2024**

## CRAG, Adaptive-RAG

Grade documents, fall back to web search; route queries by difficulty

**2024**

## Agentic RAG

LangGraph / LlamaIndex recipes: retrieval, hallucination and answer 'graders' = IsRel / IsSup / IsUse

**2025**

## Search as a tool

Assistants decide when to search and cite; RL-trained search-and-reason models

**Lasting idea:** special tokens as an interface for control and self-evaluation, and test-time reranking instead of retraining.



# Team Contribution



| Student | Contribution | Score | Main work |
|---|---|---|---|
| [Student A] | [Equal / Partial] | [1.0] | [e.g. problem & motivation, slides 2–4] |
| [Student B] | [Equal / Partial] | [1.0] | [e.g. method & equations, slides 5–11] |
| [Student C] | [Equal / Partial] | [1.0] | [e.g. experiments, slides 12–14] |
| [Student D] | [Equal / Partial] | [1.0] | [e.g. critique & impact, slides 15–16] |

Scores reflect actual relative contribution (preparation, understanding, slides, presentation), as the rubric requires.

# Thank you!

## Questions?

Paper: arxiv.org/abs/2310.11511
Code and models: selfrag.github.io



